

1 True or False

- (1) The third layer of the network stack is the transport layer.

Solution: False. The network stack is as follows: L_1 = physical layer, L_2 = datalink layer, L_3 =network, L_4 =transport, L_5, L_6, L_7 = Session, Presentation, Application.

- (2) Layering in the network stack is an example of the end-to-end principle.

Solution: False. Layering is a lesson in modularity and abstraction, breaking the functions of the data plane into smaller problems. For example L_1 deals with the physical transfer of bits which can vary depending on the fibers (optic, copper) while L_2 communicates with a single API that assembles the packet from the bits delivered by L_1 's protocol

- (3) We implement the same layers in both the host and routers.

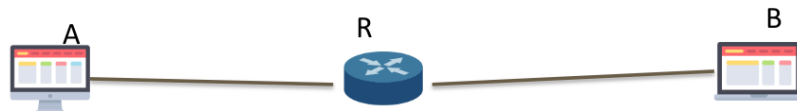
Solution: False. In the host we implement layers 1 through 7 but in a router we implement layers 1 through 3

- (4) The operating system (OS) supports logical ports while a router only has physical ports

Solution: True. A host's OS can create a logical port when using a sockets API in the transport layer. A router doesn't have an operating system and thus doesn't create logical ports for making connections using sockets.

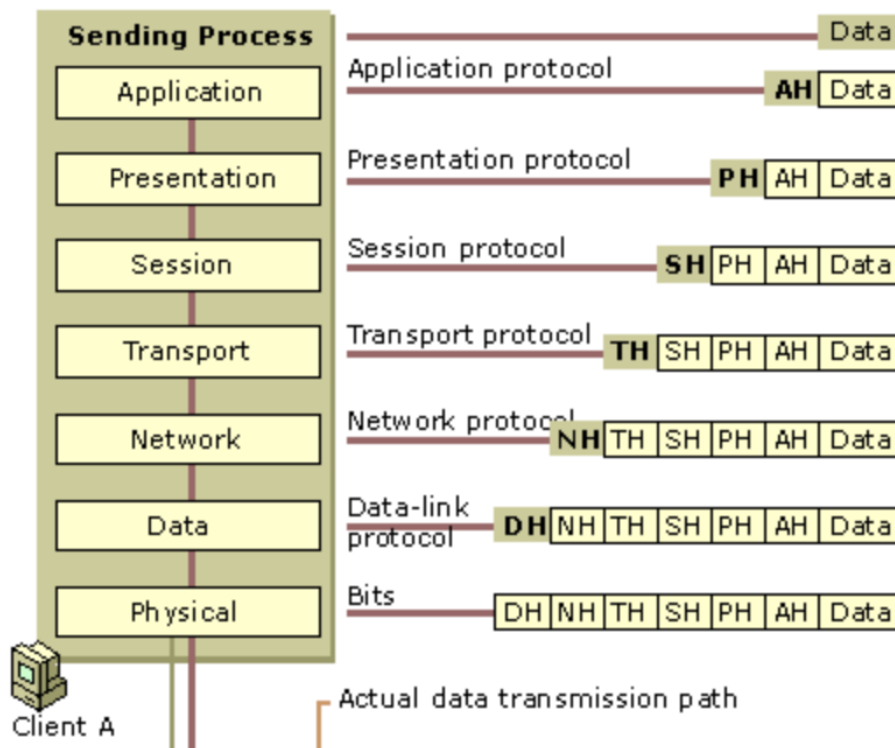
2 Network Stack

Consider host A that sends a packet to host B in the following diagram. In this problem will track what happens to the headers of the packet as it moves through the network.



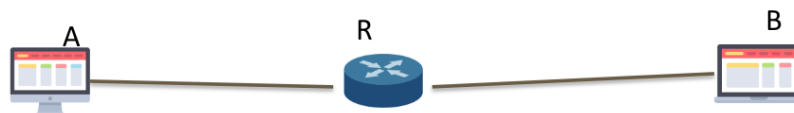
- (1) What is the network stack? **Solution:** The network stack is a design decision that handles communication of packets on the data plane. Each layer of the network stack abstracts a different part of the communication process.
- (2) Does a switch need layers 4 and 5? Why or why not? **Solution:** In the modern day internet, it would be unnecessary since a router doesn't host an application. [Tangent: perhaps layers 4 and 5 could become relevant if we wanted more secure routers.]
- (3) Host A's packet is now ready to be sent across a link. Which headers does the packet have and what is the order?

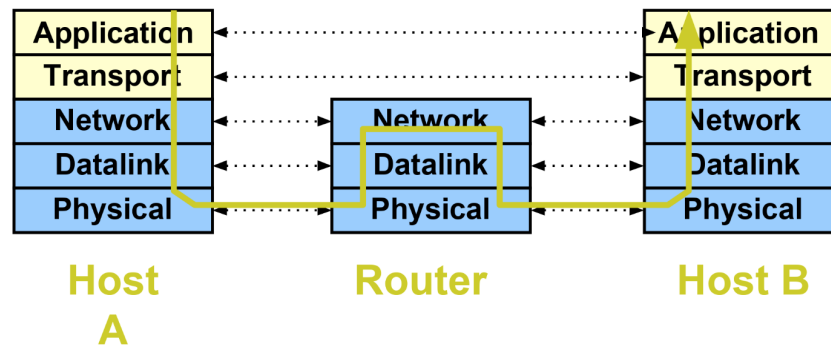
Solution: When the packet is physically about to be sent, the order of the headers from outermost to innermost will be $L_1, L_2, \dots$, up to L_7



- (4) When Host A's packet arrives at a router as bits, how do the headers of the packet change?

Solution: $L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_3 \rightarrow L_2 \rightarrow L_1$



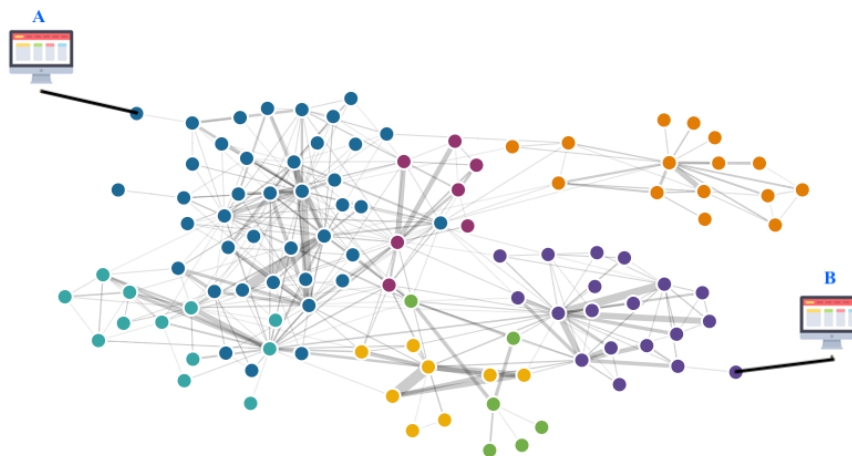


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3 Designing the Data Plane

Consider designing the internet from first principles. We'll go over how to create a design by considering how to solve the problem that lead to the modern day internet.

For this problem, assume we have many hosts communicating with each other over the network.



- (1) What problem the internet is trying to solve?

Solution: The problem the internet solved was how to have two processes on different hosts have best-effort communication with each other. The internet allows these two processes to communicate data at a distance.

- (2) What are the fundamental constraints hosts face when communicating data over a network whose resources are shared with other hosts? Create a list of the fundamental constraints of the internet. (We'll be going over this list of constraints as we progress through the semester.)

Solution:

Constraints that lead to the modern day internet

1. Traffic
2. Reliability

3. Autonomous Services
4. Routing
5. Name Services
6. Scalability

Other constraints that weren't considered when the internet was designed (and have lead to some problems)

7. Security
8. Service Classes (best effort performance, priority Routing, application based performance)

- (3) In the modern day internet, what are the physical components in a network where two hosts are communicating?

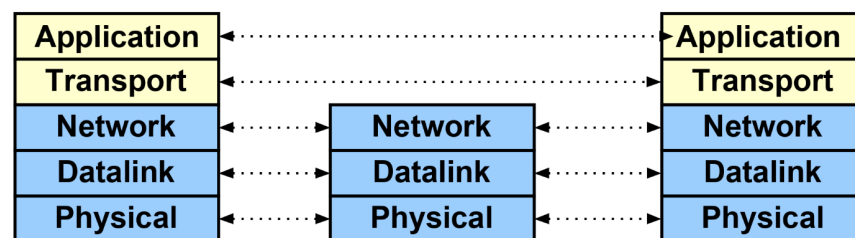
Solution: Hosts, routers, switches, links.

- (4) Focusing on routing on the data plane, how would you break up communication into smaller tasks (Note this is for a best effort design for the internet)?

Solution:

1. Putting physical bits on the wire
2. Create packets from the bits
3. Deliver packets to hosts on the same network
4. Deliver packets to hosts on different networks
5. Deliver packets reliably to hosts
6. Have the host use the data somehow

- (5) Recall the network stack of the internet. Where in the network stack should reliability, routing, and security be implemented? Consider designing functionality in only a single layer and where the state of each flow should be shared. (Note: these considerations lead to significant design decisions in the modern day internet.)



Solution:

Reliability in the modern day internet: reliability should be implemented in the Transport Layer.

Routing in the modern day internet: routing should be implemented in the Network Layer.

Security in the modern day internet: security should be implemented in the Application Layer.